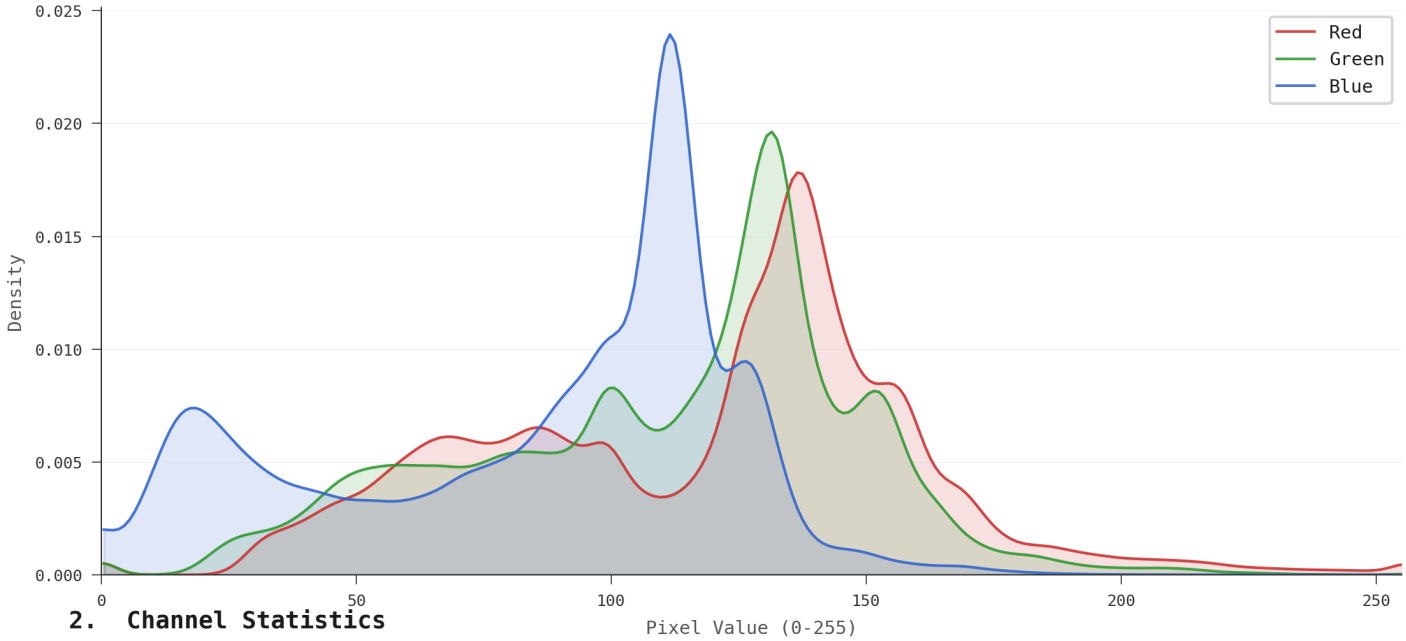
Reference Data Sheet: KODIM07

Subject: Hibiscus and Shuttered Window

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	117.5590	41.0769	1687.3098	-0.1489	-0.0814	0	255
Green	110.7168	37.6093	1414.4593	-0.2555	-0.3645	0	255
Blue	85.3725	39.2547	1540.9309	-0.7214	-0.5796	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8330	0.7634
G	0.8330	1.0000	0.9084
B	0.7634	0.9084	1.0000

Pairwise Pearson Correlation Coefficients

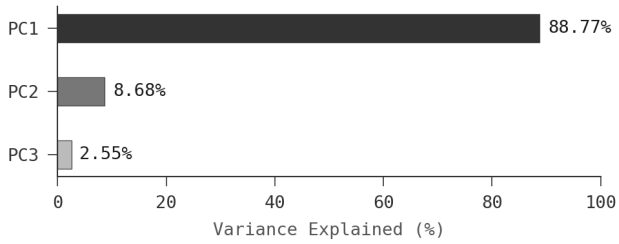
r(R-G)	=	**0.832978**
r(R-B)	=	**0.763437**
r(G-B)	=	**0.908388**
Avg |r|	=	**0.834934**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1687.3141} & 1286.8483 & 1231.0152 \\
1286.8483 & \mathbf{1414.4629} & 1341.0937 \\
1231.0152 & 1341.0937 & \mathbf{1540.9348}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM07

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	**-0.5903**	-0.5658	-0.5757	4121.3682
PC2	-0.7823	0.2253	0.5807	402.8998
PC3	0.1989	-0.7931	0.5756	118.4439

6. Derived Metrics

Condition Number

34.80

(l1/l3) PC1/PC2: 10.23 | PC2/PC3: 3.40

Blue Channel Independence

11.4%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Red

Loading: -0.5903 | Balanced

Classification: Weakly One-Dimensional (PC1 = 88.77%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.968958	0.953892	0.961425
Green	0.963485	0.940606	0.952046
Blue	0.966136	0.947753	0.956945

8. Average Local Variance (3x3)

Channel	Value
Red	117.9877
Green	124.3319
Blue	120.8266

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 34.80 | PC1: 88.77%

Correlation: G–B pair leads at r = 0.908388 | R–G: 0.832978 | R–B: 0.763437

Blue Channel: 11.4% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.956805 | Avg local variance: 121.0487

Eigenvector: Balanced | R (-0.5903), G (-0.5658), B (-0.5757) near-equal loading

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."